

(Given)

(1) States (hidden)

 $S_1 = \text{Acoustic transition}$ $S_2 = \text{steady state}$ (2) observations $O = (A, B, A)$ (3) Initial probabilities (π)

$$\pi(S_1) = 0.8, \quad \pi(S_2) = 0.2$$

(4) Transition Matrix (A) $P(\text{State}_1 \rightarrow \text{State}_2)$

From \ to	S_1	S_2
S_1	0.6	0.4
S_2	0.3	0.7

(5) Emission Matrix (B)

State	$P(A)$	$P(B)$
S_1	0.7	0.3
S_2	0.4	0.6

(Required)

(1) Number of paths = N^T

$N = 2$ (States)

$T = 3$ (length of sequence)

$$\text{then } N^T = 2^3 = 8$$

there are 8 possible paths

(2) List all paths

• A path = sequence of hidden states over time

[List all combinations of (1 & 2) of length 3]

1. 111

2. 112

3. 121

4. 122

5. 211

6. 212

7. 222

8. 221

(3) Compute path (S_1, S_1, S_1)

Given 1. observation: $O = (A, B, A)$
 2. path $Q = (S_1, S_1, S_1)$

Formula:

$$P(O, Q) = \pi(q_1) \cdot B(q_1, O_1) \cdot (A q_1 \rightarrow q_2) \cdot B(q_2, O_2) \\ \times A(q_2 \rightarrow q_3) \cdot B(q_3, O_3)$$

$$\pi(S_1) = 0.8 \rightarrow 1. \text{ initial}$$

$$B(q_1, O_1) = 0.7 \quad 2. \text{ Emission at } t=1 \quad (S_1, A)$$

$$A(q_1 \rightarrow q_1) = 0.6 \quad 3. \text{ transition } S_1 \rightarrow S_1$$

$$B(q_1 \rightarrow O_2) = 0.3 \quad 4. \text{ emission } P(q_1 \rightarrow B)$$

$$A(q_1 \rightarrow q_1) = 0.6 \quad 5. \text{ transition } P(S_1 \rightarrow S_1)$$

$$B(q_1 \rightarrow O_3) = 0.7 \quad 6. \text{ emission } P(q_1 \rightarrow A)$$

Multiply them all

$$p = \overset{\text{initial}}{\pi(s_1)} \times \overset{\text{transition}}{p(A|s_1)} \times \overset{\text{emission}}{p(s_1 \rightarrow e_1)} \times p(B|s_1) \times$$

$$p(s_1 \rightarrow s_1) \times p(A|s_1) = 0.042336$$

Step (4) Total probability

Sum the probability of all paths

$$p(0) = \sum p(0, Q)$$

given Sum of 6 paths = 0.0542

$$p(0) = 0.0542 + 0.042336 = 0.096536$$

$$p(0) = 0.096536$$

(5) Number of Paths when $T=100$

$$N^T = 2^{100}$$